

Validation against hidden ground truth

source: outputs/runs/run-20260922T041239Z-demo/module_b/validation_report.json

```
{
  "validation_version": "1.0",
  "run_dir": "outputs/runs/run-20260922T041239Z-demo",
  "module_a": {
    "truth_pairs": 95,
    "predicted_pairs": 123,
    "true_positive": 94,
    "false_positive": 29,
    "false_negative": 1,
    "precision": 0.7642,
    "recall": 0.9895,
    "refusal_ratio": 0.0956,
    "note": "Truth pairs are same-lineage adjacent versions; multi-candidate ambiguities
queued for review count as neither TP nor FP."
  },
  "module_b": {
    "snapshots": 1037,
    "test_snapshots": 208,
    "outcomes": {
      "payment": 539,
      "still_open": 488,
      "censored": 2,
      "close_no_payment": 8
    },
    "models": {
      "B0_age_stage_baseline": {
        "status": "FITTED",
        "snap_brier": 0.5144259569702185,
        "snap_log_loss": 0.7357284660871263,
        "n_snapshots": 206,
        "n_abstain": 0
      },
      "B1_discrete_multinomial": {
        "status": "FITTED",
        "snap_brier": 0.4618050188966311,
        "snap_log_loss": 0.6692132603100102,
        "n_snapshots": 206,
        "n_abstain": 0
      },
      "C1_quality_augmented": {
        "status": "FITTED",
        "snap_brier": 0.4615831802291266,
        "snap_log_loss": 0.6682848894134452,
        "n_snapshots": 206,
        "n_abstain": 0
      }
    }
  },
  "ablations": {
    "quality_features": {
      "B1_snap_log_loss": 0.6692132603100102,
      "C1_snap_log_loss": 0.6682848894134452,

```

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    "delta": -0.0009,
    "interpretation": "quality features add no incremental value on the locked test set"
  },
  "candidate_vs_baseline": {
    "B0_snap_log_loss": 0.7357284660871263,
    "B1_snap_log_loss": 0.6692132603100102,
    "relative_improvement": 0.0904
  }
},
"limitations": {
  "demo_only": "All figures come from synthetic data (seed 42). No research claim transfers to real operations.",
  "censored_share": 2,
  "close_class_rarity": 8,
  "landmark_fixed": "landmark=7d and H=30d are demo defaults needing business confirmation (unknowns U03/U04).",
  "leakage_audit": "T10/T13/T14/T15 enforced: as-of truncation, forbidden fields, lineage-grouped splits, train-only preprocessing."
}
}
```